3. $ye^{\frac{x}{y}}dx = \left(xe^{\frac{x}{y}} + y\right)dy$

4. $2xydx + \left(x^2 + 2y^2\right)dy = 0$

5. $\left(y^2 - 2xy\right)dx = \left(x^2 - 2xy\right)dy$

6. $x\frac{dy}{dx} = y - x\cos^2\left(\frac{y}{x}\right)$

7. $\left(1 + 3e^{\frac{y}{x}}\right)dy + 3e^{\frac{y}{x}}\left(1 - \frac{y}{x}\right)dx = 0$, given that $y = 0$ when $x = 1$

8. $\left(x^2 + y^2\right)dy = xy\,dx$. It is given that $y(1) = 1$ and $y(x_0) = e$. Find the value of x_0.

10.7 First Order Linear Differential Equations

A **first order differential equation** of the form

$$\frac{dy}{dx} + Py = Q. \qquad \ldots (1)$$

where P and Q are functions of x only. Here no product of y and its derivative $\frac{dy}{dx}$ occur and the dependent variable y and its derivative with respect to independent variable x occurs only in the first degree.

To integrate (1), let us consider the homogeneous equation $\frac{dy}{dx} + Py = 0$. ...(2)

The equation (2) can be integrated as follows:

Separating the variables, $\frac{dy}{y} = -Pdx$.

On integration, we get $ye^{\int Pdx} = C$.

Now, $\frac{d}{dx}\left(ye^{\int Pdx}\right) = e^{\int Pdx}\frac{dy}{dx} + y.Pe^{\int Pdx}$

$= e^{\int Pdx}\left(\frac{dy}{dx} + Py\right) = Qe^{\int Pdx}$ ... (3) (using (1))

Integrating both sides of (3) with respect to x, we get the solution of the given differential equation as

$$ye^{\int Pdx} = \int Qe^{\int Pdx}dx + C.$$

Here $e^{\int Pdx}$ is known as the **integrating factor (I.F.)** of (1).

Remarks

1. The solution of linear differential equation is
   $y \times (I.F) = \int Q(I.F)dx + C$, where C is an arbitrary constant.
2. In the integrating factor $e^{\int Pdx}$, P is the coefficient of y in the differential equation provided the coefficient of $\frac{dy}{dx}$ is unity.

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